SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	AI&ML	**Date:**	

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: "connected", "connecting", "connection", design a morphological analysis pipeline for a document indexing system.
Apply rules to decompose each word into root and suffix components.
Differentiate between inflectional and derivational forms within the dataset.
Normalize all variants to a common base form for efficient retrieval.
Determine the parsed structure and normalized output for each word.

Answer

Step 1: Morphological Analysis Pipeline

1. Input Word
2. Tokenization
3. Identify Root (Stem)
4. Separate Suffix
5. Classify as Inflectional or Derivational
6. Normalize to Base Form (Lemma)
7. Store in Index

Step 2: Decompose Each Word

Word	Root	Suffix	Type
connected	connect	-ed	Inflectional
connecting	connect	-ing	Inflectional
connection	connect	-ion	Derivational

Step 3: Inflectional vs Derivational

- Inflectional Forms
  - connected → connect + -ed (past tense)
  - connecting → connect + -ing (present participle)
- Derivational Form
  - connection → connect + -ion (changes verb into noun)

Step 4: Normalization (Lemmatization)

All words are converted to the common base form:

Original Word	Normalized Form
connected	connect
connecting	connect
connection	connect

Step 5: Parsed Structure

```
connected
├── Root: connect
└── Suffix: -ed (Inflectional)

connecting
├── Root: connect
└── Suffix: -ing (Inflectional)

connection
├── Root: connect
└── Suffix: -ion (Derivational)
```

2. Given the input words: "unhappy", "happiness", "happily", design a morphological parsing module for sentiment analysis.
Identify prefixes, suffixes, and base forms using rule-based decomposition.
Ensure semantic consistency across derived word forms.
Classify each transformation as inflectional or derivational.
Produce the root and morphological breakdown for each word.

Answer
Step 1: Morphological Parsing Module
1. Input Word
2. Identify Prefix (if any)
3. Identify Base Form (Root)
4. Identify Suffix (if any)
5. Classify Transformation
6. Normalize to Base Form
7. Use Base Form for

Sentiment Analysis Step 2: Rule-

Based Decomposition

Word	Prefix	Root (Base Form)	Suffix	Type
unhappy	un-	happy	—	Derivational
happiness	—	happy	-ness	Derivational
happily	—	happy	-ly	Derivational

Step 3: Semantic Consistency
All words are related to the base word happy.
- unhappy → opposite meaning of *happy* (negative sentiment)
- happiness → state of being *happy* (positive sentiment)
- happily → describes an action done in a *happy* manner (positive sentiment)

Step 4: Classification
- un- → Derivational Prefix (changes meaning)
- -ness → Derivational Suffix (adjective → noun)
- -ly → Derivational Suffix (adjective →

adverb) No inflectional forms are present.

Step 5: Morphological Breakdown
```
unhappy
├── Prefix: un-
├── Root: happy
└── Type: Derivational

happiness
├── Root: happy
└── Suffix: -ness (Derivational)

happily
├── Root: happy
└── Suffix: -ly (Derivational)
```

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.
Apply morphological rules to strip affixes and identify the base form.
Handle both grammatical variations and word formation changes.
Ensure consistent root extraction across all forms.
Determine the stem and classification for each input word.
Answer

Step 1: Stemming-Based Preprocessing Module

1. Input Word
2. Identify Prefix/Suffix
3. Remove Affixes (Stemming)
4. Extract Stem
5. Classify as Inflectional or Derivational
6. Store Stem for Search Index

Step 2: Morphological Rules

- Remove -ed → Past tense → play
- Remove -ing → Present participle → play
- Remove -er → Forms a noun from a verb →

play Step 3: Word Analysis

Word	Stem	Affix Removed	Classification
played	play	-ed	Inflectional
player	play	-er	Derivational
playing	play	-ing	Inflectional

Step 4: Morphological Breakdown

```
played
├── Stem: play
└── Suffix: -ed (Inflectional)
player
├── Stem: play
└── Suffix: -er (Derivational)
playing
├── Stem: play
└── Suffix: -ing (Inflectional)
```

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.
Apply state transitions to represent suffix transformations and irregular forms in verb inflection.
Differentiate between regular and irregular inflectional patterns within the dataset.
Ensure consistent mapping of all forms to a common root representation.
Determine the state transitions, morphological breakdown, and normalized output for each word.
Answer

Step 1: Finite-State Morphological Parsing Module

1. Start State (S0) → Read input word.
2. Identify Root → Detect root write.
3. Apply Transition → Remove suffix or recognize irregular form.
4. Classify → Regular or irregular inflection.
5. Normalize → Convert all forms to the base form write.
6. Store → Index the normalized form.

Step 2: State
Transitions S0 (Input)

├── writes ──remove "-s"───► S1 (write)

├── writing ─remove "-ing"──► S1 (write)

└── written ─irregular form─► S1 (write)

Step 3: Morphological Breakdown

Word	Root	Suffix/Change	Pattern	Classification
writes	write	-s	Regular	Inflectional
writing	write	-ing	Regular	Inflectional
written	write	-en(irregularpastparticiple)	Irregular	Inflectional

Step 4: Regular vs Irregular Patterns

- Regular Inflection
    - writes → write + -s
    - writing → write + -ing
- Irregular Inflection
    - written → irregular past participle of write (not formed by a simple suffix

rule) Step 5: Normalized Output

InputWord	MorphologicalBreakdown	NormalizedOutput
writes	write+s	write
writing	write+ing	write
written	write+en(irregular)	write

Conclusion

The finite-state morphological parser identifies write as the common root. It uses state transitions to process -s and -ing as regular inflectional suffixes and recognizes written as an irregular inflectional form. All variants are normalized to the base form write for consistent text processing and retrieval

5. **Given the input words: "relational", "relation", "relate", design a Porter Stemmer-based preprocessing module for document retrieval.**
Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.
Handle variations in suffix patterns while preserving semantic consistency.
Ensure uniform root extraction across all derived forms.
Determine the stemming steps, intermediate forms, and final stem for each input word.
Answer

Step 1: Porter Stemmer Preprocessing Module

1. Input Word
2. Identify Derivational Suffix
3. Apply Porter Stemming Rules
4. Remove Suffix Step-by-Step
5. Extract Common Stem

Word	Stemming Steps	Intermediate Form	Final Stem
relational	relational → remove -ational → relate → remove -e	relat	relat
relation	relation → remove -ion	relat	relat
relate	relate → remove -e	relat	relat

6. Store Stem for

Document Retrieval Step 2:

Stemming Process

Step 3: Morphological Breakdown

Word	Root	Suffix Removed	Type
relational	relate	-ational	Derivational
relation	relate	-ion	Derivational
relate	relate	-e	Base word (stemming)

Step 4: Final Stems

Input Word	Intermediate Form	Final Stem
relational	relate	relat
relation	relat	relat
relate	relat	relat

Step 5: Porter Stemmer Flow

relational
↓ remove -ational
relate
↓ remove -e
relat

relation
↓ remove -ion
relat

relate
↓ remove -e

relat

Conclusion

The Porter Stemmer removes derivational suffixes (-ational, -ion, -e) to produce the common stem relat. This ensures all related words are indexed under the same stem, improving document retrieval efficiency and maintaining semantic consistency.

Rubrics for Calculation

Numerical Problems					
Criteria	**Weighta g e**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Proble m Underst a nding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understandin g ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formu l a with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solutio n Proce s s	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculati o n	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpreta tion of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretatio n with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	**Problem Understandin g**	**Method Selection**	**Solution Process**	**Accuracy of Calculatio n**	**Interpretatio n of Results**	**Sub. Tota l**	**Total %**
1							
2							
3							
4							
5							

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________